

for keeping track of settings that apply across the entire program. If you can avoid using a global variable, that's a sign you probably should avoid using one. But "global variables are bad" is an oversimplified opinion.

Myth: Comments Are Unnecessary

Bad comments are indeed worse than no comments at all. A comment with outdated or misleading information creates more work for the programmer instead of a better understanding. But this potential problem is sometimes used to proclaim that *all* comments are bad. The argument asserts that every comment should be replaced with more readable code, to the point that programs should have no comments at all.

Comments are written in English (or whatever language the programmer speaks), allowing them to convey information to a degree that variable, function, and class names cannot. But writing concise, effective comments is hard. Comments, like code, require rewrites and multiple iterations to get right. We understand the code we write immediately after writing it, so writing comments seems like pointless extra work. As a result, programmers are primed to accept the "comments are unnecessary" viewpoint.

The far more common experience is that programs have too few or no comments rather than too many or misleading comments. Rejecting comments is like saying, "Flying across the Atlantic Ocean in a passenger jet is only 99.999991 percent safe, so I'm going to swim across it instead."

Chapter 10 has more information about how to write effective comments.

Summary

Code smells indicate that there might be a better way to write your code. They don't necessarily require a change, but they should make you take another look. The most common code smell is duplicate code, which can signal an opportunity to place code inside a function or loop. This ensures future code changes will need to be made in only one place. Other code smells include magic numbers, which are unexplained values in your code that can be replaced by constants with descriptive names. Similarly,